

## List

- List items are **ordered**, **changeable**, and **allow duplicate** values. and written with **square brackets[]** and List items can be of any data type. **Ex:-** list = ["abc", 34, True, 40, "male"]
- **length:-** using **len** function Ex:- print(len(thislist))
- **Check if Item Exists:-** using **in** Ex:- if "apple" in thislist:

### Add List Items:-

- **Append:-** End of the list **Ex:-** thislist.append("orange")
- **Insert:-** Specified index **Ex:-** thislist.insert(1, "orange")
- **Extend** Add any iterable **Ex:-** thislist.extend(sets)

```
thislist = ["apple", "banana", "cherry"]

# Append Method:- Add an item to the **end of the list**, use the **append()** method.
thislist.append("orange")
print(thislist) # Output:- ['apple', 'banana', 'cherry', 'orange']

# Insert method:- The **insert()** method inserts an item at the ***specified index***.
thislist.insert(1, "orange")
print(thislist) #Output:- ['apple', 'orange', 'banana', 'cherry']

# Extend method:- The **extend()** method does not have to append lists, you can
**add any iterable** object (list, tuples, sets, dictionaries etc.).
sets = ("mango", "pineapple", "papaya")
thislist.extend(sets)
print(thislist) #Output:- ['apple', 'banana', 'cherry', 'mango', 'pineapple', 'papaya']
```